Prednisone and Prednisolone

Mechanism of action

- Prednisone is a prodrug that is metabolised to prednisolone in the liver.
- Prednisolone is a corticosteroid with anti-inflammatory and immunosuppressant actions. It inhibits the production of inflammatory mediators, including prostaglandins and leukotrienes, resulting in a reduction in the inflammatory and immune response.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Bronchospasm associated with asthma or COPD.
- ✓ Croup.
- ✓ Minor allergy associated with rash.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Age less than five years with asthma. Steroids do not usually have a role in children aged less than five years because they do not generally alter the course of their asthma exacerbation. However, a steroid is indicated if the child has a clear history of asthma and has previously received oral steroids.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. However, there is significant clinical experience with steroids and they appear to be safe. A steroid should be administered if there is bronchospasm, but should be withheld if the clinical problem is minor, for example rash or itch.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Prednisone and Prednisolone

Dosage

- 40 mg for an adult.
- See the paediatric drug dose tables for a child.
- If the patient is already taking prednisone:
  - Administer an additional full dose if the patient is taking a dose that is lower than that described in these CPGs. If the patient is not transported by ambulance to a medical facility, advise the patient to discontinue their usual prednisone, take the prednisone supplied by personnel and have their treatment reviewed in primary care (preferably by their own GP) within two days.
  - Do not administer an additional dose if the patient is taking a dose equal to or higher than that described in these CPGs. If the patient is not transported by ambulance to a medical facility, advise the patient to continue taking their usual prednisone and have their treatment reviewed by a doctor (preferably their own GP) within two days.

Administration

- Administer PO.
- Prednisone tablets are very bitter. Do not crush prednisone tablets because this may cause vomiting.
- The tablets may be divided. The tablet does not always break evenly, and this is not of clinical importance. The larger piece of the divided tablet should be chosen for administration.
- Patients aged greater than or equal to ten years should usually be administered prednisone tablets but may be administered the same dose of prednisolone syrup.
- Patients aged less than ten years should usually be administered prednisolone syrup but may be administered the same dose of prednisone tablets.
- Always ask a parent (or guardian) of a young child if the child can swallow tablets.

Common adverse effects

- Fatigue.
- Sodium and water retention. This may worsen hypertension and heart failure, but is usually only of clinical significance with prolonged dosing.
- Gastrointestinal reflux.

Prednisone and Prednisolone

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 24 hours.

Usual preparation

- 5 mg prednisone tablets.
- 20 mg prednisone tablets.
- Prednisolone syrup containing 5 mg/ml.

Pharmacokinetics

- Prednisone and prednisolone are predominantly metabolised by the liver.
- Prednisone is a prodrug that is metabolised to prednisolone in the liver. A dose of prednisone provides slightly less steroid equivalence than the same dose of prednisolone, but this is not clinically significant.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.

Providing the patient with prednisone

- If the patient is aged greater than or equal to 12 years and is not transported, additional prednisone should be provided to them by ambulance personnel, unless the patient already has an asthma/COPD action plan and their own supply of steroid.
- Provide the patient with an additional four days (four x 40 mg) of prednisone. Advise the patient and any carers, including in writing, that the patient:
  - Takes 40 mg (two 20 mg tablets) at approximately the same time each day, for the next four days until finished, in addition to their other usual medicines.

Prednisone and Prednisolone

» Should be reviewed in primary care for a review of their treatment within two days.

More Information

Version history

16 June 2026 1.1.0.1
Update: Treatment guidance updated.

28 October 2024 1.0.5
Update: Indications updated.

22 November 2022 1.0.1
Update: Prednisone pack information.